

Unlock the Potential of R for Data Analytics

Since the last decade, the R programming language has assumed importance as the most important tool for computational statistics, visualisation and data science. R is being used more and more to solve the most complex problems in computational biology, actuarial science and quantitative marketing.

Since data volumes are increasing exponentially, storage has become increasingly complex. Rudimentary tips and techniques have become obsolete and no longer result in improved efficiency. Currently, complex statistical and probabilistic approaches have become the *de facto* standard for major IT companies to harness deep insights from Big Data. In this context, R is one of the best environments for mathematical and graphical analysis on data sets. Major IT players like IBM, SAS, Statsoft, etc, are increasingly integrating R into their products.

R code: A walk through

Numerical computation: It is very easy to use R for basic analytical tasks. The following code creates a sample data of claims made in the year 2013 and subsequent claims that turned out to be fraudulent for a hypothetical insurance

company *ABC Insurance Inc.* Then, the code performs some simple commonly used statistical tasks like calculating extended mean (number of values greater than 29 upon total number of values), standard deviation, degree of correlation, etc. Finally, the last statement performs a test to check whether the means of two input variables are equal. A null hypothesis results if the means are equal; else, an alternative hypothesis emerges. The p-value in the result shows off the probability of obtaining a test (here, it is extremely small as there is a huge difference between the means):

```
> claim_2013 <- c(30,20,40,12,13,17,28,33,27,22,31,16);
> fraud_2013 <- c(8,4,11,3,1,2,11,17,12,10,8,7);
> summary(claim_2013);

> mean(claim_2013>29);
      Output: [1] 0.3333333
> sd(claim_2013);
      Output: [1] 8.764166
> cor(claim_2013, fraud_2013);
      Output: [1] 0.7729799
> table(fraud_2013, claim_2013);
> t.test(claim_2013, fraud_2013);
```

Probabilistic distribution:

R provides an over-abundance of probabilistic function support. It is the scenario that decides the domain of probabilistic applicability.

Exponential distribution: This is an estimator of the occurrence of an event in a given time frame, where all other events are independent and occur at a constant rate. Assume that a project's stub processing rate is $\rho = 1/5$ per sec. Then, the probability that the stub is completed in less than two seconds is:

```
> pexp(2, rate=1/5);
```

Binomial distribution: This is an estimator of the exact probability of an occurrence. If the probability of a program failing is 0.3, then the following code computes the

